Image-based simulation of morphogen gradient formation during zebrafish epiboly

Justina Stark[1, 2, 3], Rohit Krishnan Harish[1, 4, 5], Michael Brand[1, 4, 5], Ivo F. Sbalzarini[1, 2, 3, 5]

1 Technische Universität Dresden / 2 Center for Systems Biology Dresden / 3 MPI-CBG Dresden / 4 Center for Regenerative Therapies Dresden / 5 Cluster of Excellence Physics of Life

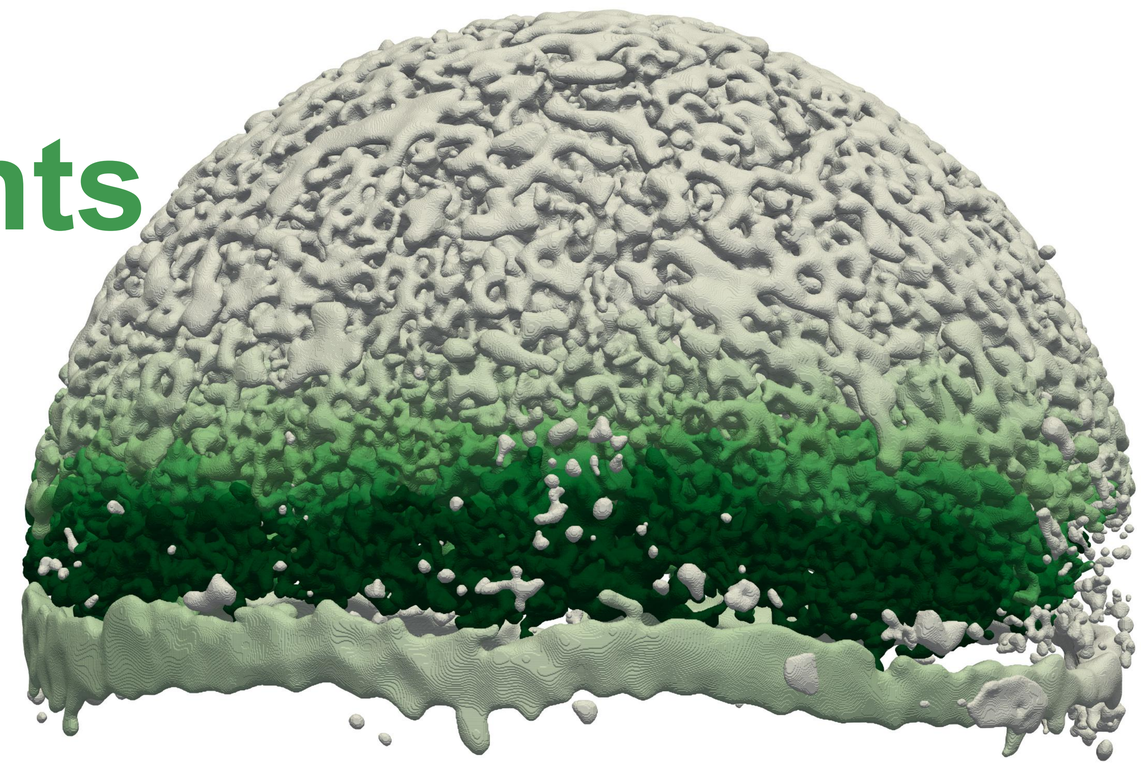
jstark@mpi-cbg.de

IB-COMBRYOS

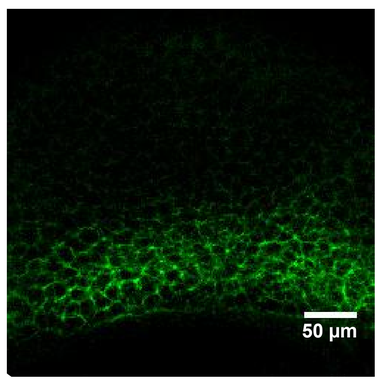
help understand how

morphogen gradients

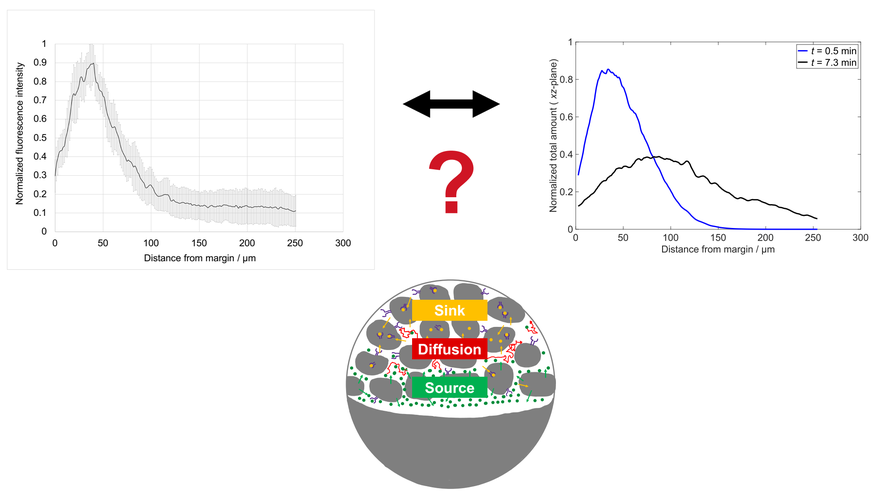
form in the complex shapes of growing embryos



Light microscopy images / volumes[1]



Validation and testing hypotheses[6]



$$\frac{\partial c(\boldsymbol{x},t)}{\partial t} = k_{\text{prod}}(\boldsymbol{x},t)\cdot c_{\text{prod}}(\boldsymbol{x},t) + D\cdot\Delta c(\boldsymbol{x},t) - k_{\text{degr}}(\boldsymbol{x},t)\cdot c(\boldsymbol{x},t)$$

Simulation of morphogen gradient formation in image-based geometry

Generic image-based simulation pipeline

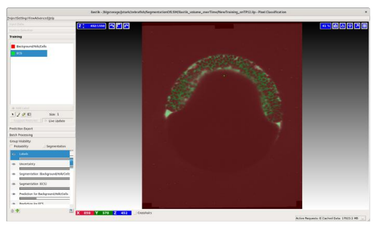
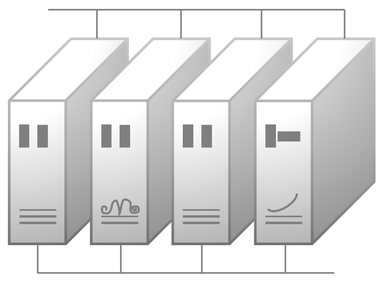
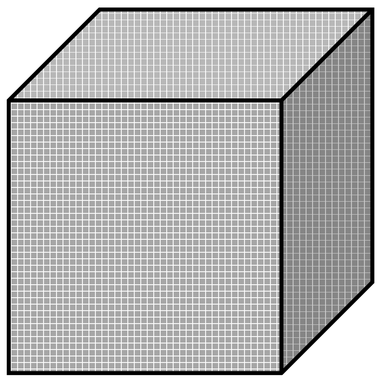


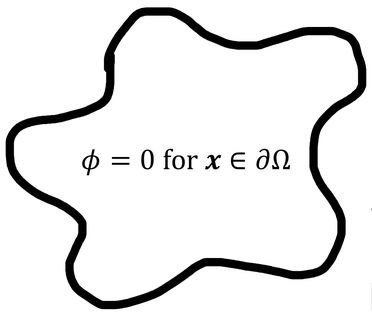
Image segmentation using ilastik[2]



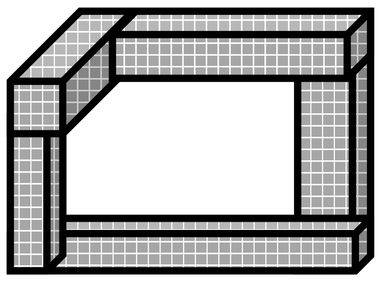
Parallelization using OpenFPM[5]



Transfer of binary mask to computational grid



Surface representation with level-set methods[3]



Reduction of computational and memory cost using sparse grids on GPU[4]

Acknowledgements

We thank our collaboration partners Rohit Krishnan Harish and Michael Brand for the microscopy images, the experimental data, and the discussions.
I thank my colleagues (the MOSAIC group) and Ivo Sbalzarini for their steady support and just everything.

TECHNISCHE UNIVERSITÄT DRESDEN

CENTER FOR SYSTEMS BIOLOGY DRESDEN

CBG Max Planck Institute of Molecular Cell Biology and Genetics

References

[1] R. K. Harish, M. Gupta, D. Zöller, H. Hartmann, A. Gheisari, A. Machate, S. Hans and M. Brand, "Real-time monitoring of endogenous Fgf8a gradient attests to its role as a morphogen during zebrafish gastrulation," *bioRxiv preprint*, 2022.
[2] C. Sommer, C. Strähle, U. Köthe, and F. A. Hamprecht, "ilastik: Interactive Learning and Segmentation Toolkit," *in Eighth IEEE ISBI*, 2011.
[3] M. Sussman and E. Fatemi, "Efficient, interface-preserving level set redistancing algorithm and its application to interfacial incompressible fluid flow," *SIAM J. Sci. Comput.*, 1999.
[4] P. Incardona, T. Bianucci, and I. F. Sbalzarini, "Distributed Sparse Block Grids on GPUs", *ISC High Performance*, 2021.
[5] P. Incardona, A. Leo, Y. Zaluzhnyi, R. Ramaswamy, and I. F. Sbalzarini, "OpenFPM: A scalable open framework for particle and particle-mesh codes on parallel computers," *Comput. Phys. Commun.*, 2019.
[6] S. R. Yu, M. Burkhardt, M. Nowak, J. Ries, Z. Petrásek, S. Scholpp, P. Schwille and M. Brand, "Fgf8 morphogen gradient forms by a source-sink mechanism with freely diffusing molecules," *Nature*, 2009.